

Goals:

- Definition and examples of neighborhoods;
- A neighborhood may or may not be open;
- A set is open iff it is a neighborhood of all its points;
- Definition of interior point and interior of a set as the set of all interior points;
- Characterization of the interior of a set A as the largest open set contained in A ;
- Characterization of open sets as sets coinciding with their interior.

Review:

- Families of sets and their unions;
- Basis for a topology;
- Cofinite topology, lower limit topology.

Definition 1. Let $(X, \mathcal{T})$ be a topological space and $x \in X$. A **neighborhood** of x is a subset U of X such that there exists $A \in \mathcal{T}$ such that $x \in A \subseteq U$. The family of all neighborhoods of x is called the **neighborhood system** of x and is denoted by $\mathcal{N}(x)$.

Exercise 1 Consider $\mathbb{R}$ with the Euclidean topology and $x \in \mathbb{R}$.

- i) Give an example of a neighborhood of x that is open and one that is not open.
- ii) Give an example of a subset of $\mathbb{R}$ containing x that is not a neighborhood of x

Proposition 1. Let X be a topological space and $A \subseteq X$. We have that A is open if and only if for all $x \in A$ we have $A \in \mathcal{N}(x)$

Definition 2. Let X be a topological space, $A \subseteq X$, and $x \in X$.

- We say that x is an **interior point** of A if $A \in \mathcal{N}(x)$, i.e. if there exists an open set U such that $x \in U \subseteq A$.
- The **interior** of A is defined as

$$\mathring{A} = \text{int}(A) = \{x \in X \mid x \text{ is an interior point of } A\}.$$

Observation: Let $(X, \mathcal{T})$ be a topological space and $A \subseteq X$. Then,

- $\mathring{A} \subseteq A$.
- If $\mathcal{B}$ is a basis for $\mathcal{T}$, then $x \in \mathring{A}$ if and only if there exists $B \in \mathcal{B}$ such that $x \in B \subseteq A$.

Proposition 2. Let X be a topological space and $A \subseteq X$. We have that

$$A \text{ is open if and only if } A = \mathring{A}$$

Proposition 3. *Let X be a topological space and $A \subseteq X$. The interior of A is the largest open set contained in A . Equivalently,*

$$\mathring{A} = \bigcup_{\substack{U \text{ open} \\ U \subseteq A}} U$$

Exercise 2 For $a < b \in \mathbb{R}$ consider $A =]a, b[$, $B = [a, b[$, $C = [a, b]$, and $D = \mathbb{Q}$. Find the interior of A , B , C , D as subsets of $\mathbb{R}$ endowed with

- i) the Euclidean topology
- ii) the lower limit topology
- iii) the cofinite topology